Acids & salts

Understand solutions

1 pH & neutralisation

Acid: H^+ in water; alkali: OH^-. Below pH 7 acidic; 7 neutral; above 7 alkaline.

$H^+ + OH^- \rightarrow H_2O$. Universal indicator estimates pH; a calibrated probe gives a numerical reading.

Equal volumes need not contain equal reacting amounts.

2 Strength vs concentration

Strong acids fully ionise: HCl, HNO_3, H_2SO_4. Weak acids partly ionise: ethanoic, citric, carbonic.

Concentrated/dilutedescribes amount per volume, not ionisation.

pH falls by 1 → H^+ concentration ×10. pH 2 has 100× the H^+ of pH 4.

Predict products

3 Acid reaction patterns

Metal + acid → salt + H_2. Metal oxide/hydroxide + acid → salt + water.

Carbonate + acid → salt + water + CO_2. H_2 gives squeaky pop; CO_2 turns limewater cloudy.

4 Name & formulate salts

HCl → chloride; H_2SO_4 → sulfate; HNO_3 → nitrate. Metal supplies positive ion/name.

Balance charges: $Mg^{2+} + 2NO_3^- \rightarrow Mg(NO_3)_2$. $2Na^+ + SO_4^{2-} \rightarrow Na_2SO_4$.

$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$.

Prepare pure crystals

5 Salt practical: react & filter

Warm dilute acid gently; remove burner. Add insoluble oxide/carbonatein small portions, stirring, until excess remains.

Excess solid ensures all acid reacts. Filter off solid; salt solution is **filtrate.**

Goggles; avoid boiling/splashing acid.

6 Concentrate & crystallise

Gently evaporate some water using water bath/electric heater. Cool to form crystals.

Filter, rinse with a little cold distilled water and dry between filter papers. Do not strongly heat to dryness.

Soluble alkali cannot be filtered off: find exact volumes by titration instead.